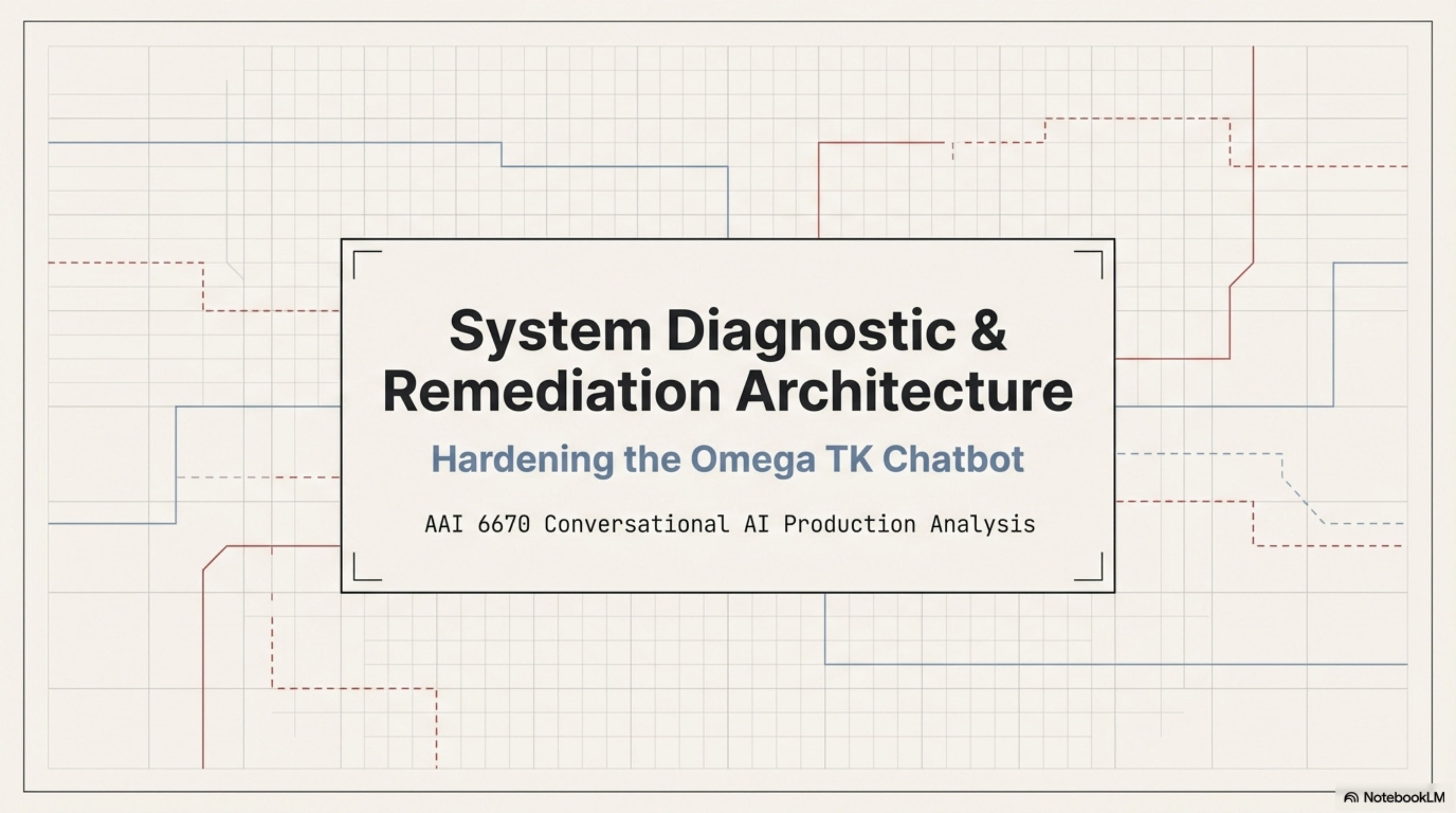




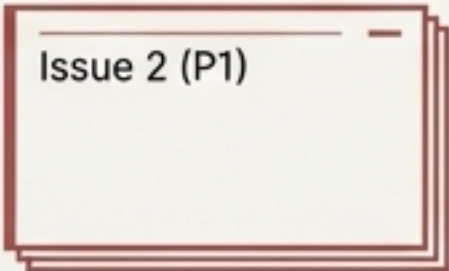
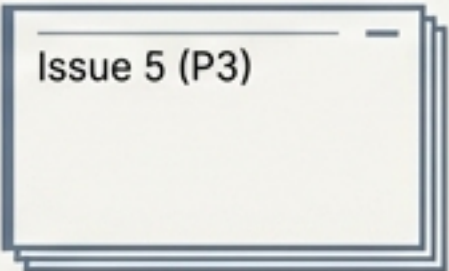
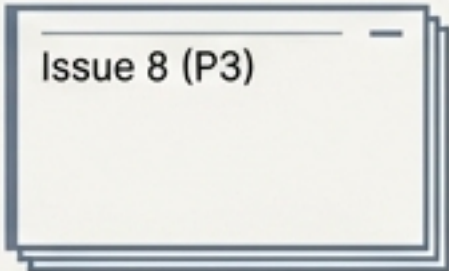
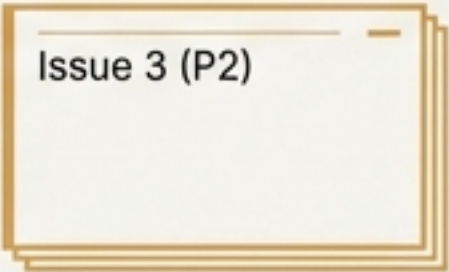
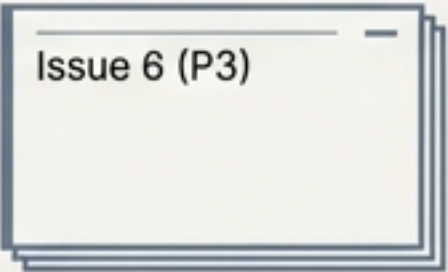
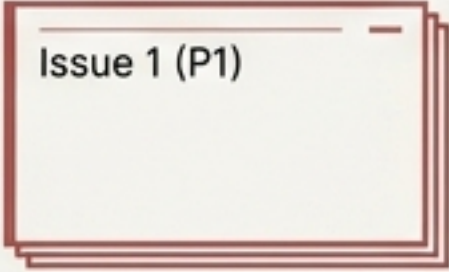
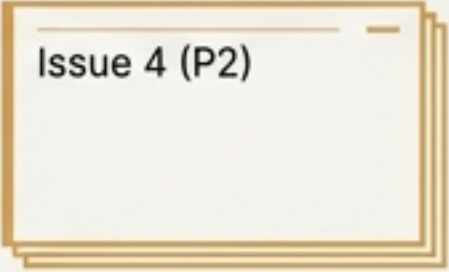
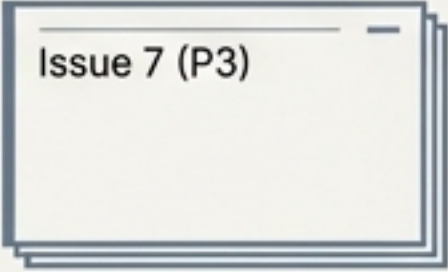
System Diagnostic & Remediation Architecture

Hardening the Omega TK Chatbot

AAI 6670 Conversational AI Production Analysis

8 Real Issues. 3 Independent Sprints. 0 Full Rewrites.

The production codebase requires targeted remediation, not a systemic overhaul. Every sprint is independently deployable.

Sprint 1: Backend Hardening + Tests	 Issue 2 (P1)	 Issue 5 (P3)	 Issue 8 (P3)
Sprint 2: Analytics Loop + Summary		 Issue 3 (P2)	 Issue 6 (P3)
Sprint 3: Knowledge Base Architecture	 Issue 1 (P1)	 Issue 4 (P2)	 Issue 7 (P3)

Prerequisite Data Infrastructure

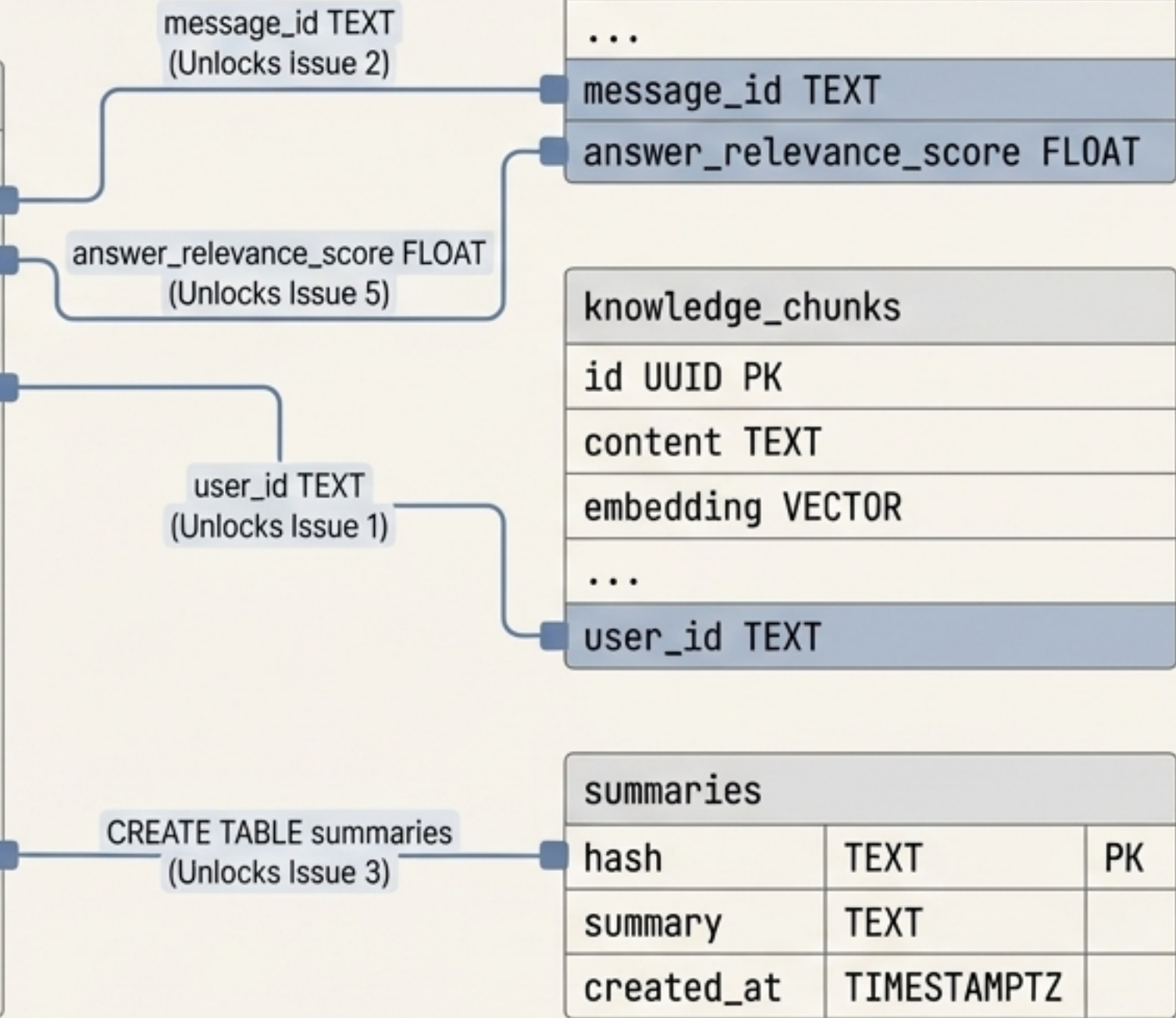
Additive, non-breaking schema additions that must run in the Supabase SQL Editor before any code deployment.

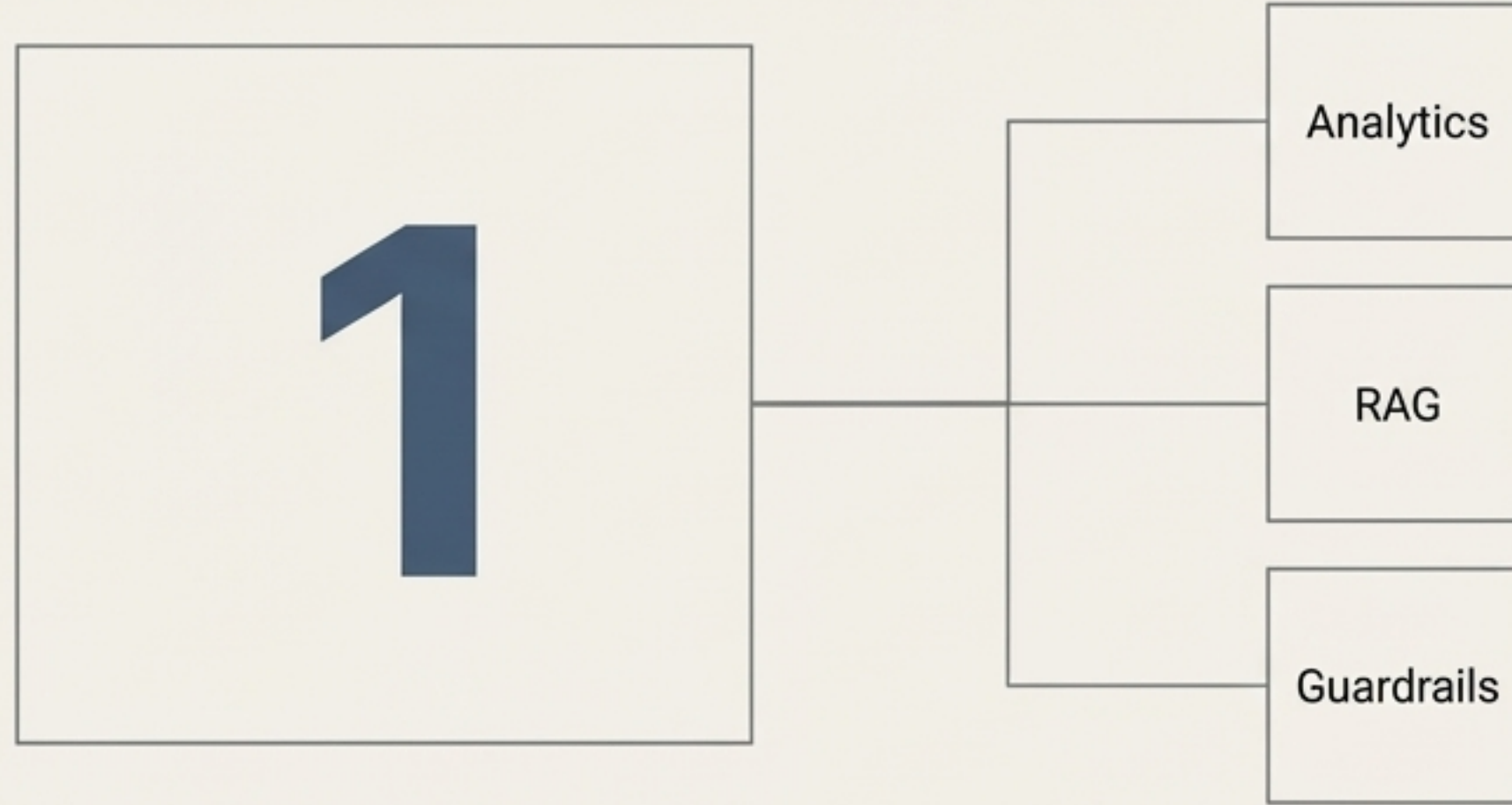
```
ALTER TABLE queries_log ADD COLUMN message_id TEXT;
ALTER TABLE queries_log ADD COLUMN answer_relevance_score FLOAT;

ALTER TABLE knowledge_chunks ADD COLUMN user_id TEXT;

CREATE INDEX IF NOT EXISTS idx_knowledge_chunks_user_id
  ON knowledge_chunks(user_id);

CREATE TABLE summaries (
  hash TEXT PRIMARY KEY,
  summary TEXT NOT NULL,
  created_at TIMESTAMPTZ DEFAULT now()
);
```





Sprint 1: Backend Hardening + Tests

Pure backend changes. No UI work. Independently deployable in a single PR.

Targets: Add 11 adversarial tests, close the analytics join gap, and track answer relevance on every query.

Closing the Analytics Join Gap (P1)

Current State



Symptom: Feedback is collected, but cannot be tied to the specific response, FAISS score, or retry count. Debugging relies on guesswork.

Target State



The Fix: Generate a UUID message_id on request. Frontend stores it, sends it with thumbs feedback. Ten lines of code in app.py.

Expanding RAG Evaluation Observability (P3)

The RAG Triad Matrix	
Current State	Sprint 1 State
Retrieval Confidence The diagram shows a large triangle representing the RAG Triad. The top vertex is labeled 'Retrieval Confidence', the bottom-left is 'Answer Relevance', and the bottom-right is 'Faithfulness'. Inside, a smaller triangle is shaded in grey and labeled 'faiss_score'. Another smaller triangle is shaded in red with diagonal lines. The 'Answer Relevance' area is mostly empty. Blindspot: Currently, hallucinations that bypass Layer 3 validation or subtly miss the prompt are invisible in analytics.	Retrieval Confidence The diagram shows the same RAG Triad structure. In addition to the 'faiss_score' (grey) and the red-shaded triangle, a new blue-shaded triangle is added, labeled 'answer_relevance_score'. This blue triangle is positioned towards the 'Answer Relevance' vertex, indicating improved visibility into that area. The Fix: Utilize the existing <code>_embed_text</code> helper to embed both original query and response. Compute cosine similarity. One embedding call per non-fallback response.

Hardening Adversarial Defenses (P3)

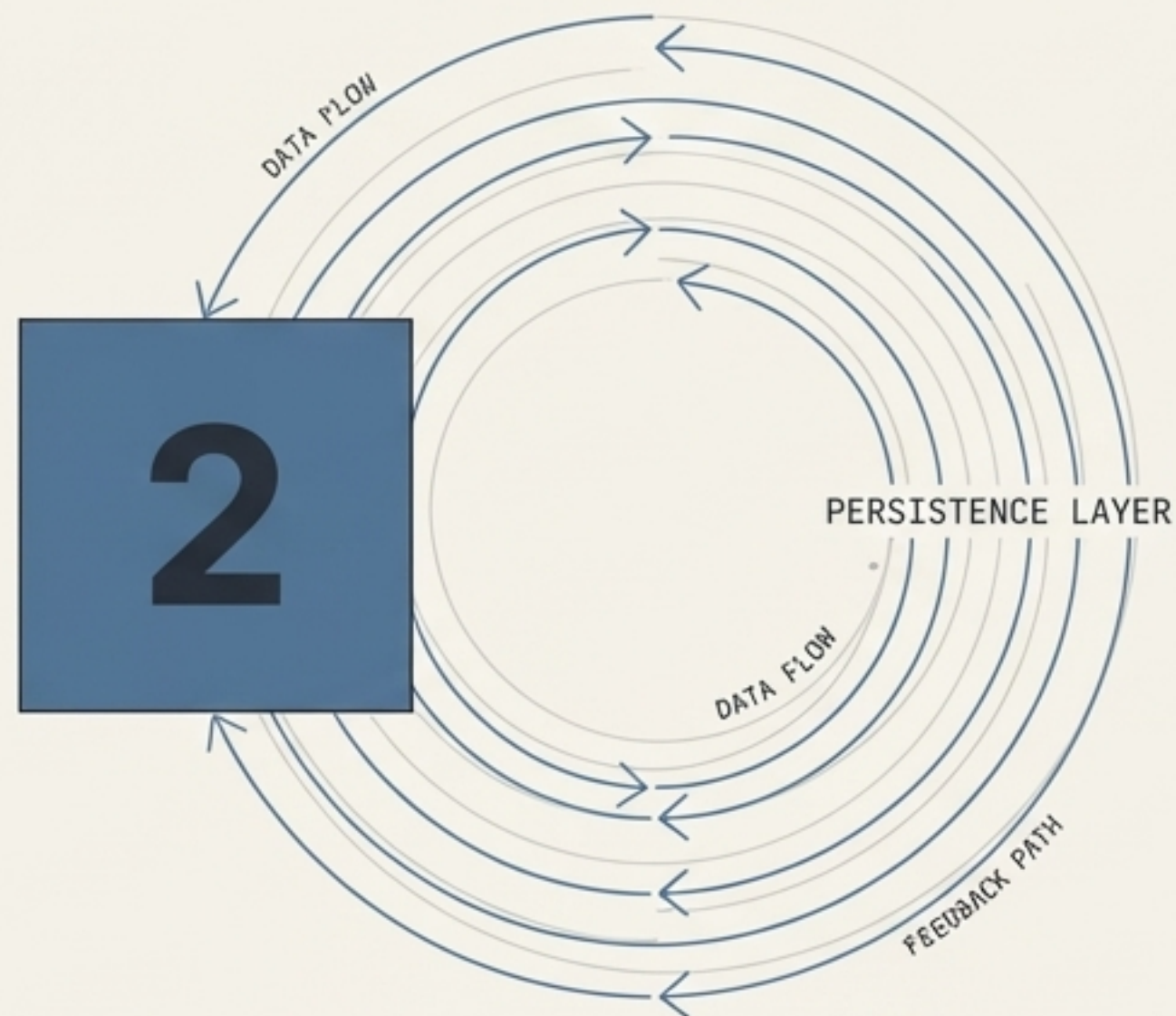
Current State



Current State: The blocklist has more words than the test suite has tests (exactly 2 tests exist). **Injection** variants and roleplay bypasses are entirely untested. Production LLM systems require 10-20% adversarial coverage.

The Fix

- Inject 5 new blocklist terms: `'no restrictions'`, `'without restrictions'`, `'developer mode'`, `'do anything now'`, `'unrestricted'`.
- Deploy `TestL1Adversarial` class with 11 new test cases.
- Target Metric: **70** tests passing (up from 59).



Sprint 2: Analytics Loop + Summary Persistence

Description: Closes the feedback loop to surface actionable data and hardens the conversation summary cache against server restarts.

Dependency: Relies on Sprint 1 data infrastructure.

Activating the Feedback Loop (P3)

timestamp	feedback_type	user_id	message_id
2023-10-27T10:15:30Z	👎 thumbs-down	user_123	msg_456
2023-10-27T10:16:45Z	👎 thumbs-down	user_124	msg_457
2023-10-27T10:18:12Z	👎 thumbs-down	user_125	
...	...	...	
...	...	...	
...	...	...	...
...	...	...	...
...	...	...	...

JOIN ON message_id

Needs Review

User Message: "The answer was incorrect and confusing."

Intent: "Product Information"

Latency: 450ms

Lowrating: 👎

Symptom: A helpful rate percentage is displayed, but measuring for the sake of measuring is useless. The evaluation loop exists in name only.

The Fix: Query the 5 most recent thumbs-down ratings, join to queries_log via the new message_id, and render a "Needs Review" card in AnalyticsDashboard.jsx.

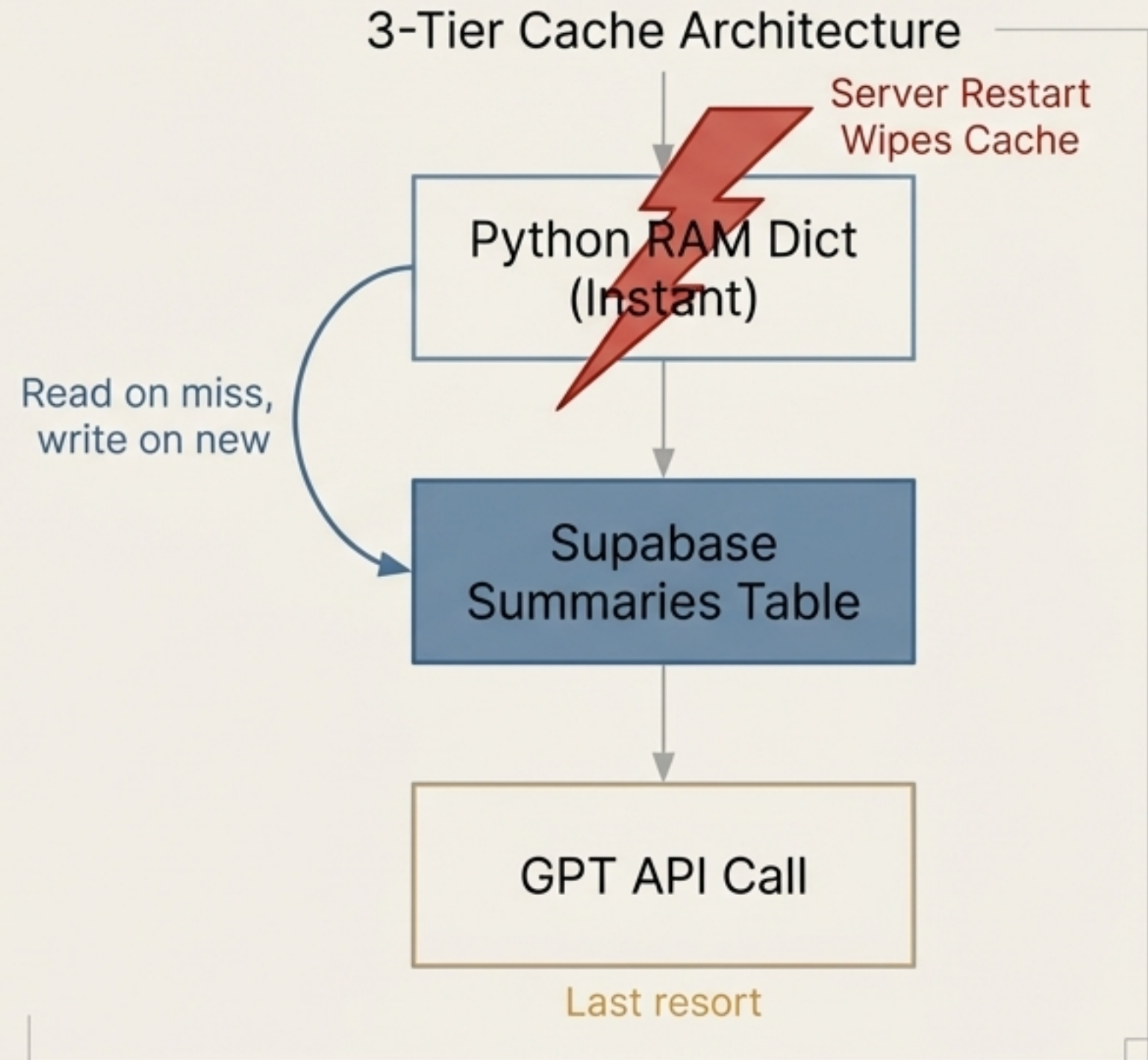
Persistent Conversation Summaries (P2)

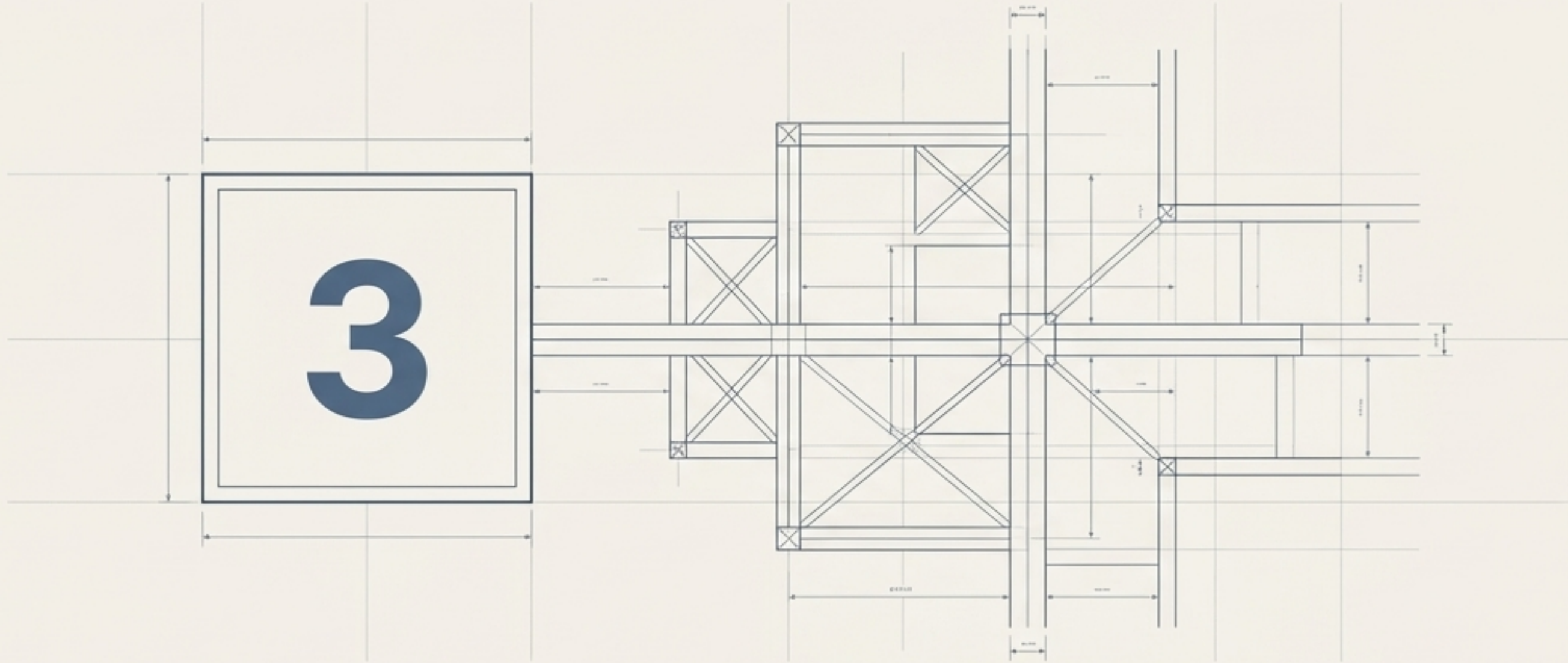
Symptom:

Summary cache lives entirely in Python RAM. Server restarts silently wipe it, forcing expensive, latent re-summarizations via GPT for ongoing long conversations.

The Fix:

Upgrade `_get_summary()` to check Supabase on an RAM cache miss before calling OpenAI.





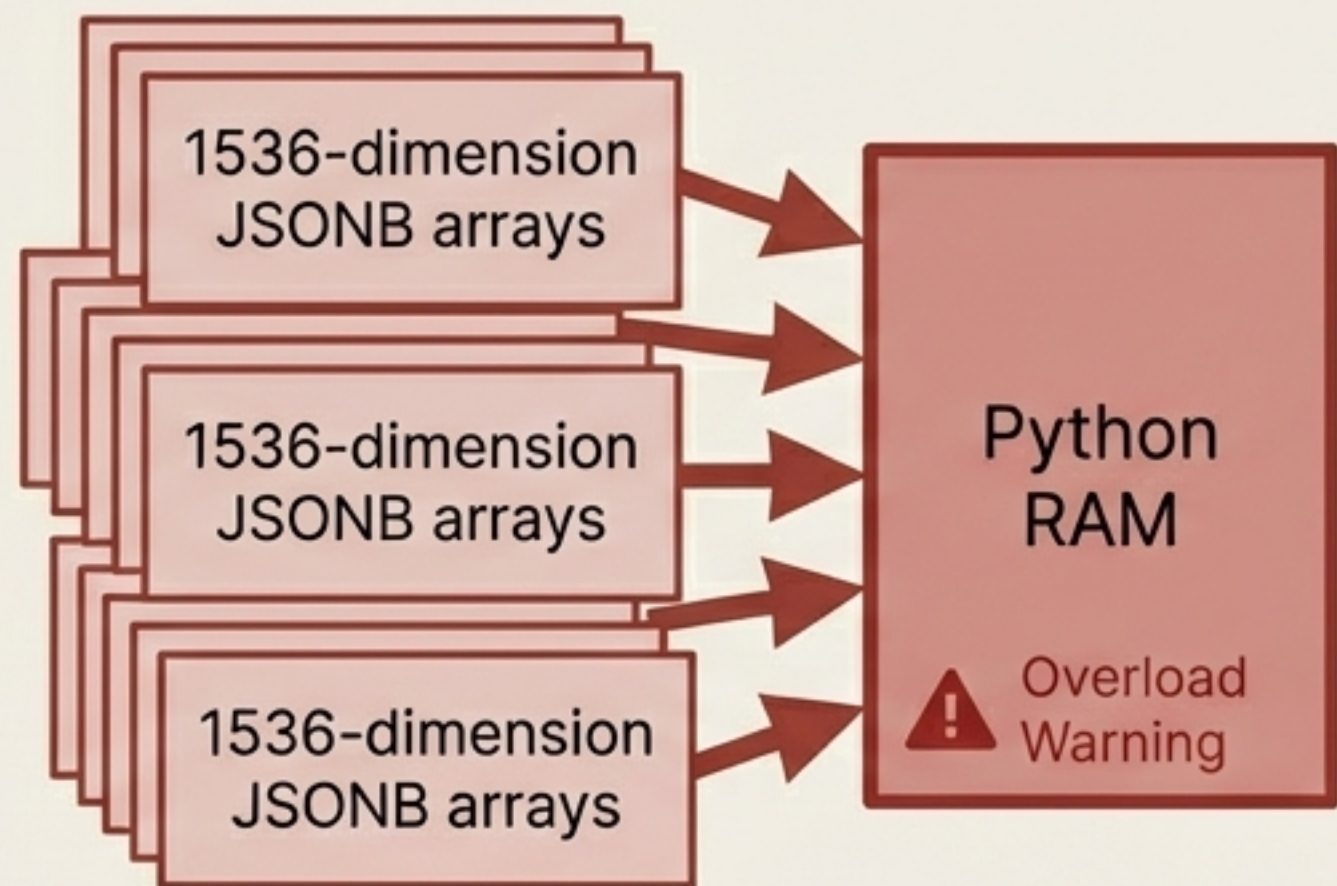
Sprint 3: Knowledge Base Architecture

Description: The largest sprint. Introduces persistent browser identity and applies safety caps to knowledge retrieval.

Impact: Fixes the only issue causing permanent user data loss.

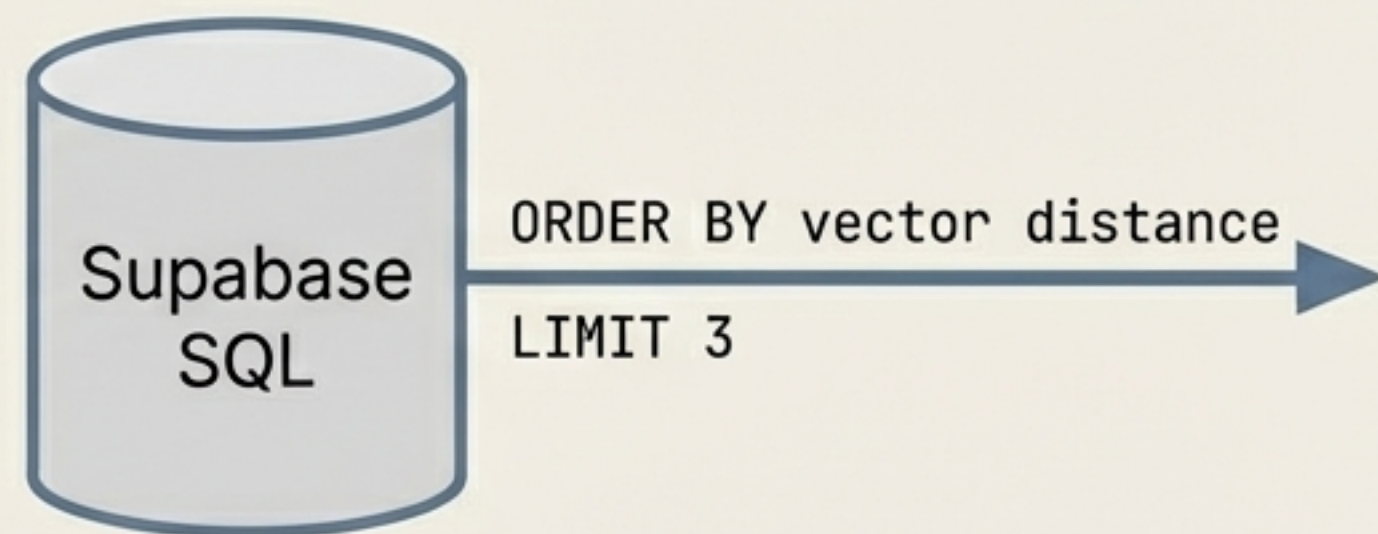
Mitigating RAM Overload (P2)

Broken



Symptom: The system currently fetches every chunk into Python memory to run a cosine similarity loop. Fine for 5 chunks; causes latency spikes and silent failures at 50.

Fixed



The Fix: Immediate 500-row safety cap and >100 chunk log warning. Permanent fix delegates similarity search to SQL via pgvector.

Establishing Persistent Identity (P3)

Current State



Symptom: Sessions are completely anonymous (UUIDs in localStorage). Clearing cache or switching devices results in total identity loss, blocking V3 — Auth and Knowledge Base isolation.

Sprint 3 State



The Fix: Introduce `omega_user_id` as a persistent browser identity that survives 'New Chat' clicks. A pragmatic stepping stone to full email/password Supabase Auth in V3.

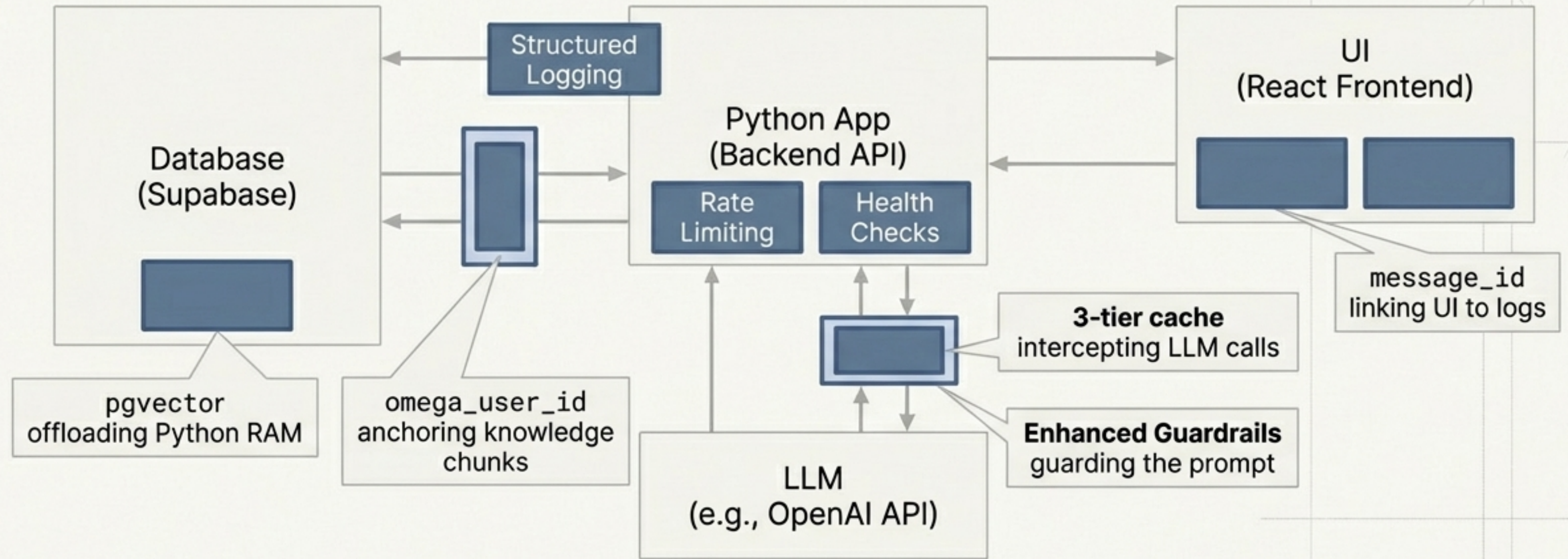
Deep Dive: The “Lost Knowledge” Phenomenon (P1)

Context: 1536-dim vectors live in Supabase, not RAM. They aren't deleted; they are orphaned by session rotation.

User Action	Environment	Local State	Resulting System State
Upload → Refresh	Local Dev	Missing Config	Appears Empty (✗)
Upload → Refresh	Production	session_id intact	Survives (✓)
Upload → New Chat	Any	New session_id	Invisible / Orphaned in DB (✗)
Upload → New Device	Any	Different session_id	Invisible (✗)

The Fix: Tag all uploads with the new omega_user_id instead of ephemeral session IDs.

The Production-Ready Architecture



Takeaway: Through 3 highly targeted, independently deployable sprints, the system transitions from a fragile prototype to a resilient, stateful, and fully observable production asset—with zero core rewrites and all 70 tests passing.